

Jeremy Hsu

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EDUCATION

Pennsylvania State University

Bachelor of Science, Computer Science (College of Engineering)

University Park, PA

Aug. 2025 – May 2028

- **Relevant Coursework:** Data Structures, Discrete Mathematics for Computer Science, Linear Algebra, Multivariable Calculus & Vector Analysis, Applied Regression Analysis

EXPERIENCE

Founder & Software Engineer

Weaver Capital LP

Jan. 2026 – Present

West Chester, PA

- Built and shipped 2 production trading systems in Python and C# executing live against broker APIs, with automated strategy synchronization, enforced risk limits, and sub-second order handling
- Designed a shared decision path between research and production to prevent model/live divergences, persisting system state across restarts with atomic writes
- Built data pipelines ingesting market and macroeconomic feeds (FRED, Yahoo, broker APIs) with Parquet caching, data-quality checks, and lookahead-safe time alignment
- Shipped Streamlit/Plotly monitoring dashboards surfacing live positions, system state, and flow-adjusted performance

Undergraduate Researcher

Locomotion in Biology and Robotics (LiBR) Lab, Penn State

Aug. 2026 – Present

University Park, PA

- Building physics simulations of multi-body contact dynamics in MuJoCo, modeling an articulated three-section snake robot navigating confined tunnel geometry to evaluate gait strategies before hardware testing
- Modeling scaling laws for computational versus mechanical cost of transport under Dr. Baxi Chong

Research Intern

Robot Perception and Navigation Group, University of Delaware

June 2024 – July 2024

Newark, DE

- Built a 2D SLAM pipeline on a TurtleBot3 using ROS 2 and Google Cartographer in C++ to map an indoor lab
- Integrated Nav2 for autonomous waypoint navigation on the resulting map
- Presented navigation and sensing results to graduate researchers and faculty

Founder & Technical Lead

FTC #16468 Green Lemons Robotics

June 2019 – Apr. 2025

West Chester, PA

- Achieved #1 OPR globally among 8,000+ teams at the 2024 FTC World Championships
- Founded and led a 14-member team across six seasons, directing recruiting, training, and technical direction
- Developed autonomous control software in Java, including vision-based navigation, odometry, and path planning

PROJECTS

Rime | *Python, pandas, NumPy, Alpaca API, Streamlit, pytest*

2025 – Present

- Built a scheduled daily allocator that computes a target portfolio from market and macro features and submits orders through a broker REST API, running autonomously since Apr. 2026
- Wrote 47 pytest tests covering the decision tree, state transitions, gating logic, and risk overlays; validated end-to-end against 13 years of historical data under walk-forward and cost-stress scenarios
- Implemented idempotent order handling, a rebalance deadband to suppress redundant trades, and atomic state persistence with a JSON-backed store so a crashed or interrupted run resumes cleanly

Kintoun | *Python, NumPy, pandas*

Aug. 2025 – Present

- Built an event-driven backtesting engine for a 15-minute opening-range breakout on NQ futures 1-minute bars
- Engineered the data layer for DST-aware US/Eastern alignment and exclusion of identified feed-degradation days
- Vectorized indicator computation in NumPy/pandas while keeping execution in an event-driven loop; simulated 1,062 trades over 2022–2025

TECHNICAL SKILLS

Languages: Python, C++, Java, C#, SQL

Robotics & Simulation: ROS 2, Cartographer SLAM, Nav2, MuJoCo, RViz2, LiDAR, OpenCV, path planning

Systems & Data: REST APIs, pytest, pandas, NumPy, SciPy, Parquet, Matplotlib, Streamlit

Tools: Git, Linux, Jupyter, NinjaTrader